Assignment: module - 5
Network Fundamentals and Building Networks

Section 1: Multiple Choice Questions

1. What is the primary function of a router in a computer network?

a) Assigning IP addresses to devices
b) Providing wireless connectivity to devices
c) Forwarding data packets between networks
d) Managing user authentication and access control

Ans: Forwarding Data Packets between the networks.

2. What is the purpose of DHCP (Dynamic Host Configuration Protocol) in a computer network?

a) Assigning static IP addresses to devices
b) Resolving domain names to IP addresses
c) Managing network traffic and congestion
d) Dynamically assigning IP addresses to devices

Ans: Dynamically assigning IP Addresses to devices

3. Which network device operates at Layer 2 (Data Link Layer) of the OSI model and forwards data packets based on MAC addresses?

a) Router
b) Switch
c) Hub
d) Repeater

Ans: Switch

4. Which network topology connects all devices in a linear fashion, with each device connected to a central cable or backbone?

a) Star
b) Bus
c) Ring
d) Mesh

Ans: Bus

Section 2: True or False

True or False: A VLAN (Virtual Local Area Network) allows network administrators to logically segment a single physical network into multiple virtual networks, each with its own broadcast domain.

True

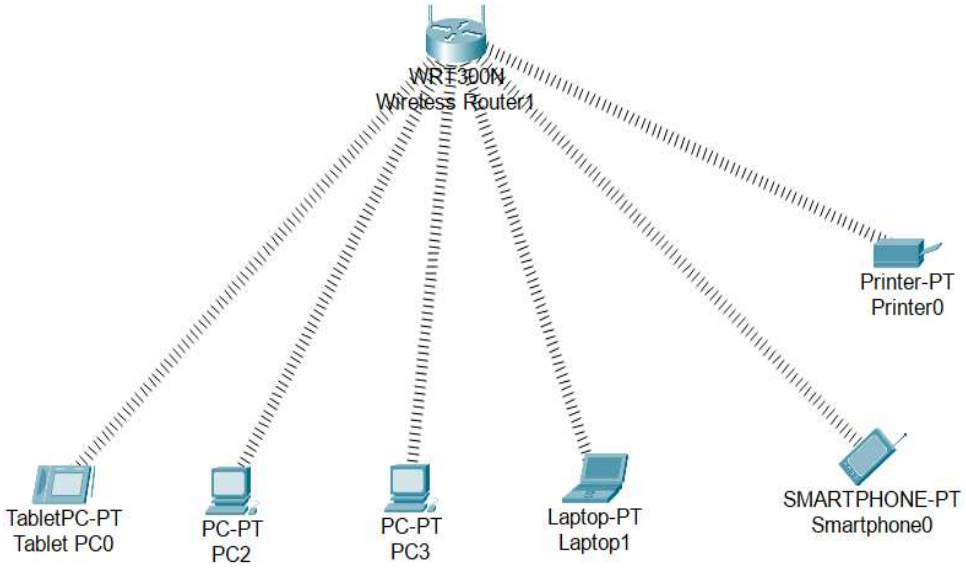
True or False: TCP (Transmission Control Protocol) is a connectionless protocol that provides reliable, ordered, and error-checked delivery of data packets over a network.

False

True or False: A firewall is a hardware or software-based security system that monitors and controls incoming and outgoing network traffic based on predetermined security rules.

True

8. Describe the steps involved in setting up a wireless network for a small office or home office (SOHO) environment.



Ans:

Select no of devices for usage requirement.
Select the wireless router (Here WRT 300N).
Select assigning of Ip Address using DHCP protocol in Wireless router.
Choose wifi password and username.
Select restricting Port number of Restricted usage.
Upgrade or update router if necessary.

Section 5:

Section 4: Practical

9. Demonstrate how to configure a router for Internet access using DHCP (Dynamic Host Configuration Protocol).

Ans:

Router

Steps:

Config mode: ip dhcp pool kunj(name)

Config-dhcp: network 192.168.29.0(Network ID) 255.255.255.0(Subnet Mask)
default-router 192.169.29.1(Router IP)
dns-server 192.168.29.254

Config mode: ip dhcp excluded-address 192.168.29.1 192.168.29.10
(Excluded IP Address Range)

10. Discuss the importance of network documentation in the context of building and managing networks.

Ans:

- Troubleshooting IP network protocols.
  - Assigning IP Addressing
  - Reserved IP Address
- Network monitoring and maintenance**.**
  - Updating firmware
  - Adding new devices
- Improves Security
- Easy for Auditing and Inspection
- Easily know to all other working in network maintainance.
- Easy Recovery in case of network down.
- Device inventory can easily managed.
- Reqiure devices is use at maximum occupancy.
-